UNIVERSITY OF WATERLOO

SYDE 556/750 – Simulating Neurobiological Systems

Test 2: Temporal Representation & Feedforward Transformations

Student ID: _____ Time: 30 minutes Total: 20 marks (plus 2 bonus)

- **Closed book.** No notes, textbook, calculator, phone, or other devices.
- The marks for each question are shown in brackets on the right.
- Write your answers in the space provided. Show your work — partial credit is given for correct reasoning.

Question 1. **[4 marks]**

You want to generate a band-limited white noise signal $x(t)$ of length T seconds with RMS σ and an upper frequency limit L (in Hz).

(a) Describe in 3–4 short steps how you would construct $x(t)$ by sampling random values in the frequency domain.

(b) What constraint must the Fourier coefficients $X(\omega)$ satisfy so that $x(t)$ comes out purely real?

(c) In the box, sketch the average $|X(\omega)|$ you would expect over many such signals. Label the axes and mark the location of the limit L.

Question 2. **[4 marks]**

The following equation is used for the LIF neuron:

$$\frac{dv}{dt} = \frac{1}{\tau_{RC}}(J - v).$$

Explain what each term in the equation represents, and what the equation as a whole describes (including what happens when v reaches threshold).

Question 3. **[4 marks]**

The post-synaptic current filter is $h(t) = c^{-1}t^n e^{-t/\tau}$ for $t \geq 0$, where c normalizes it to unit area. (a) Sketch $h(t)$ for $n = 0$ and for $n = 1$ on the same axes. (b) With $n = 0$, if you increase τ, name two things that happen to the decoded estimate $\hat{x}(t)$.

(a)	(b)

Question 4. **[4 marks]**

You want a network in which population P_1 represents x and drives population P_2 so that P_2 represents $y = 2x + 1$.

(a) What decoders do you solve for?

(b) Describe each step in the computation (there are 6) that occurs as signals flow from the input to the decoded output.

Question 5. **[4 marks]**

Consider a population of *two* LIF spiking neurons with identical parameters ($\tau_{RC} = 20$ ms, $\tau_{\text{ref}} = 2$ ms, same α and J^{bias}) but opposite encoders, $e_1 = +1$ and $e_2 = -1$.

In the box, sketch *all of the following* on the same time axis:

- the input signal $x(t)$ — a sine wave, a couple of periods long;
- the spike trains of both neurons;
- your best guess at the decoded signal $\hat{x}(t)$.

Bonus Question 1. **[2 bonus marks]**

When decoding a spike train, the optimal linear filter can be written

$$H(\omega) = \frac{\langle X(\omega)\, R^*(\omega) \rangle}{\langle |R(\omega)|^2 \rangle},$$

where $R(\omega)$ is the spike-response spectrum. In plain language, what is this filter doing, and how does it use the signal and response spectra?